

Bucephalus AI Inventory Assistant

Bucephalus is a Next.js and Office.js Excel add-in that discovers, cleans, validates, and analyzes inventory data across multi-sheet workbooks. It supports inconsistent headings, combined product/inventory tables, SKU aliases, multiple locations, currencies, units of measure, sales history, supply data, movement ledgers, LCM valuation, aging reserves, forecasting, and reorder recommendations.

All inventory calculations are performed deterministically in TypeScript. Botpress is optional and is used only to explain verified results and answer workbook questions.

Requirements

- Microsoft Excel for Mac or Windows
- Microsoft 365 account
- Node.js 22 or a supported active-LTS release
- npm
- Git
- Botpress account only if chatbot functionality is required

Installation

Clone the repository and install its dependencies:

```
git clone <YOUR-GITHUB-REPOSITORY-URL>
cd Bucephalus-AI-Inventory-Assistant
npm install
```

Start the local HTTPS development server:

```
npm run dev
```

The task pane will run at:

```
https://localhost:3000
```

Accept or trust the local development certificate if prompted.

Sideload in Excel for Mac

Copy the included `manifest.xml` file into Excel's add-in directory:

```
mkdir -p ~/Library/Containers/com.microsoft.Excel/Data/Documents/wef  
cp manifest.xml ~/Library/Containers/com.microsoft.Excel/Data/Documents/wef/
```

Restart Excel, open a workbook, and select Bucephalus AI Inventory Assistant from the Excel ribbon.

For Windows or centralized Microsoft 365 deployment, follow Microsoft's Office Add-in sideloading or administrative deployment process using the included manifest.

Using the add-in

1. Open an inventory workbook (see folder named "Test Workbooks" or choose your own).
2. Select Enterprise Workbook.
3. Click Scan Enterprise Workbook.
4. Review the proposed dataset roles, column mappings, defaults, duplicates, and warnings.
5. Correct or exclude anything that was interpreted incorrectly.
6. Confirm the warnings and run Enterprise Analysis.
7. Review the newly created Buc Worksheets.

Bucephalus never overwrites the original source worksheets. It creates normalized item, inventory, sales, supply, aging, forecast, and exception sheets.

The original fixed-table workflow remains available under Simple Table Fallback and expects an Excel table named InventoryTable.

Optional Botpress Setup

Publish a Botpress agent, configure Webchat in embedded mode, and copy its public configuration-script URL. Create `.env.local`:

`NEXT_PUBLIC_BOTPRESS_CONFIG_URL=`<https://files.bpcontent.cloud/your-config.js>

Restart the development server. After running Enterprise Analysis, use Share Full Workbook Context & Open Chat to send the current workbook context to Botpress.

Workbook data is not shared automatically. The user must approve sharing. Botpress may explain verified results but should not recalculate or override inventory quantities, valuations, forecasts, reserves, or recommendations.

Verification

Before deploying changes, run:

```
npm test  
npm run lint  
npm run build
```

The project currently passes 39 automated tests covering several enterprise workbooks, unusual headings, combined datasets, currencies, UOM conversions, duplicate sources, repeated snapshots, forecasting, reserves, and Botpress context handling.

This is a development add-in. Production use still requires hosted HTTPS, authentication, privacy review, security testing, Microsoft 365 administrative deployment, and organizational approval.